Energy Security:

Importance, Effects, and Policies in South Korea

Energy Sources

Fossil Fuel

Cost-effective
Established Infrastructure

Major Pollutant
Finite Recourse
Energy Dependence

Nuclear Energy

Energy Independence
Continuous Generation

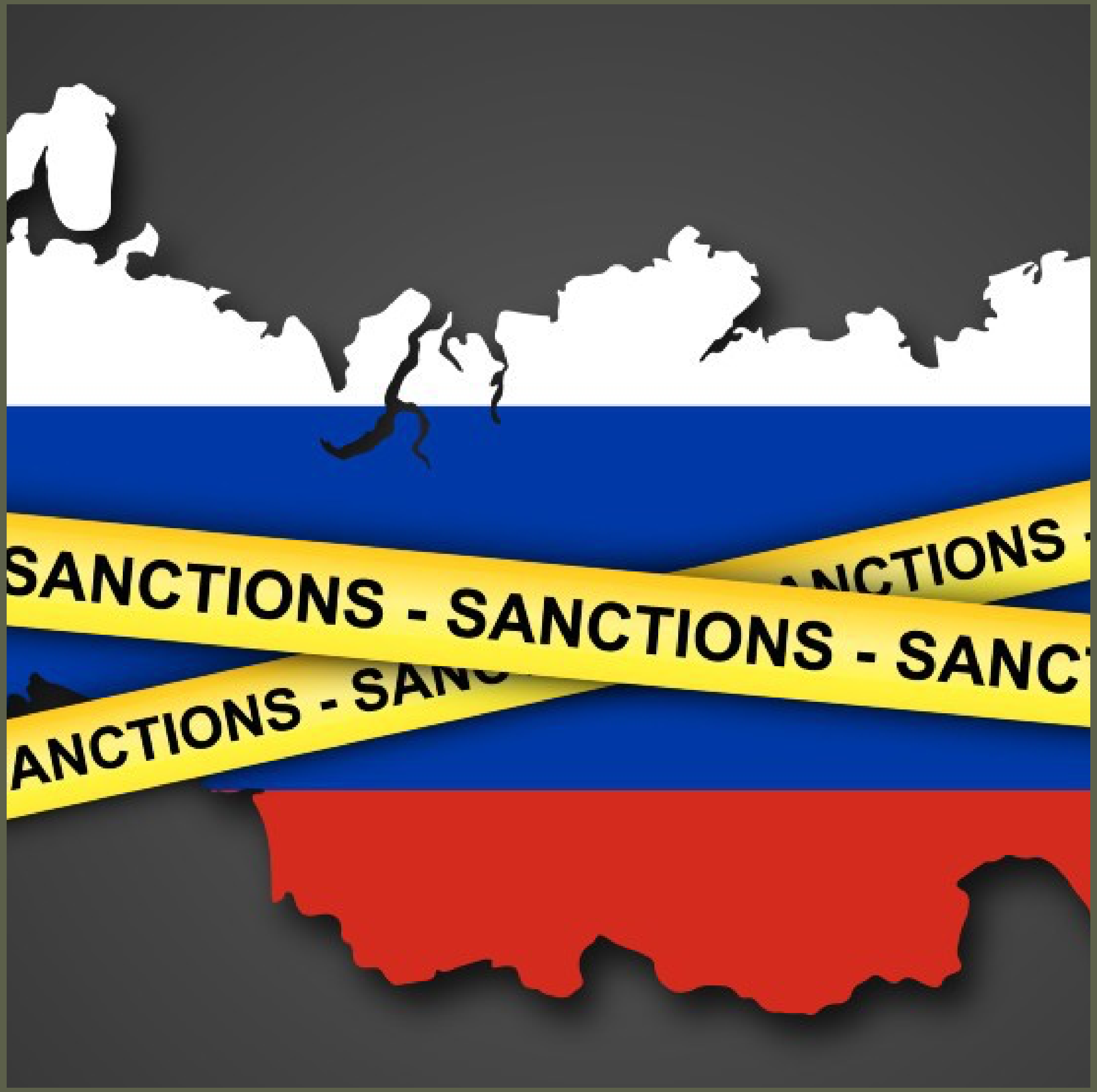
Radioactive Waste & Accidents
High Initial Cost
Decommissioning Cost

Renewable Energy

Energy Independence
Environmental Benefit

Insufficient Production
High Initial Cost

War Exposes Vulnerability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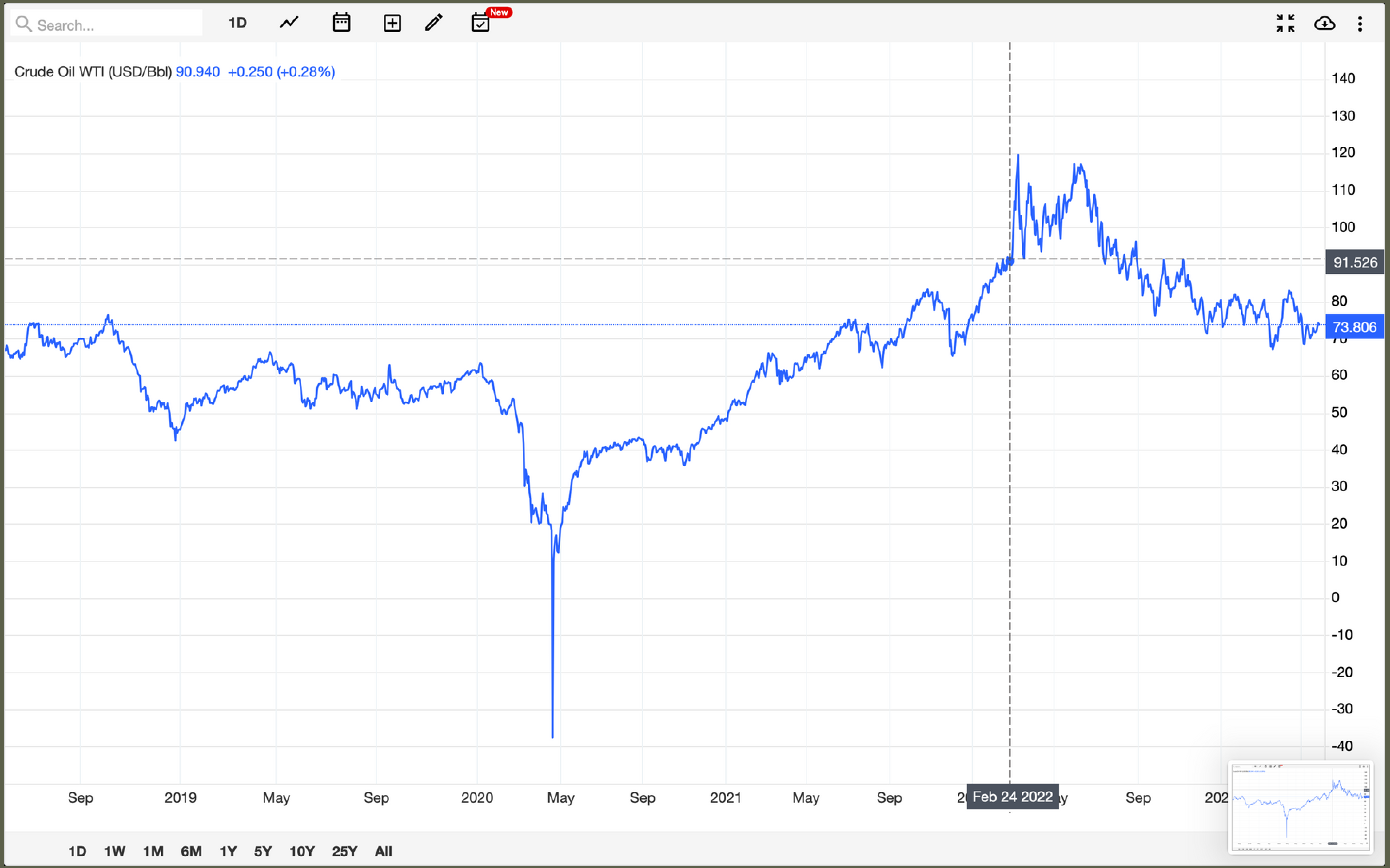
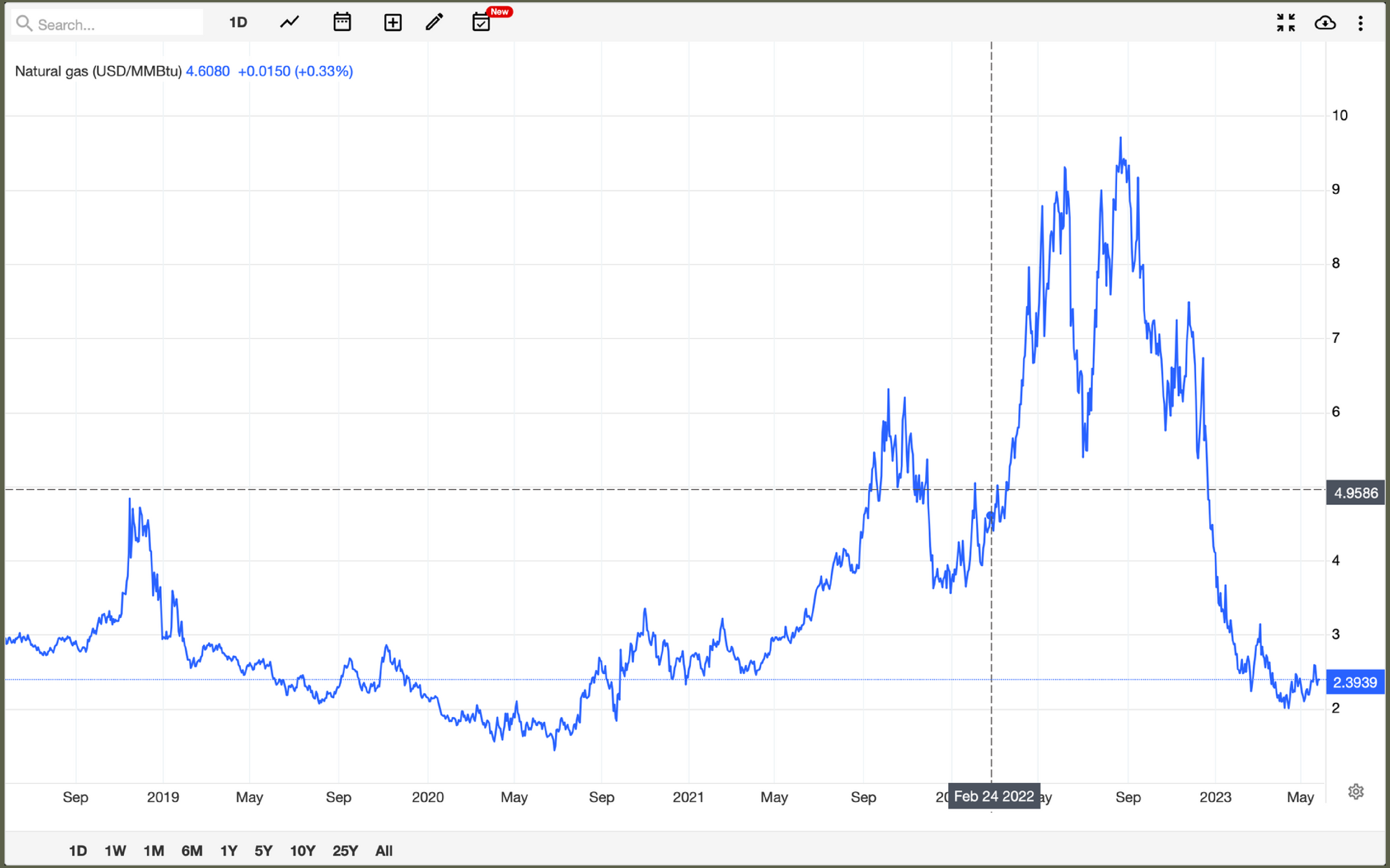
Commodity Prices Spiked

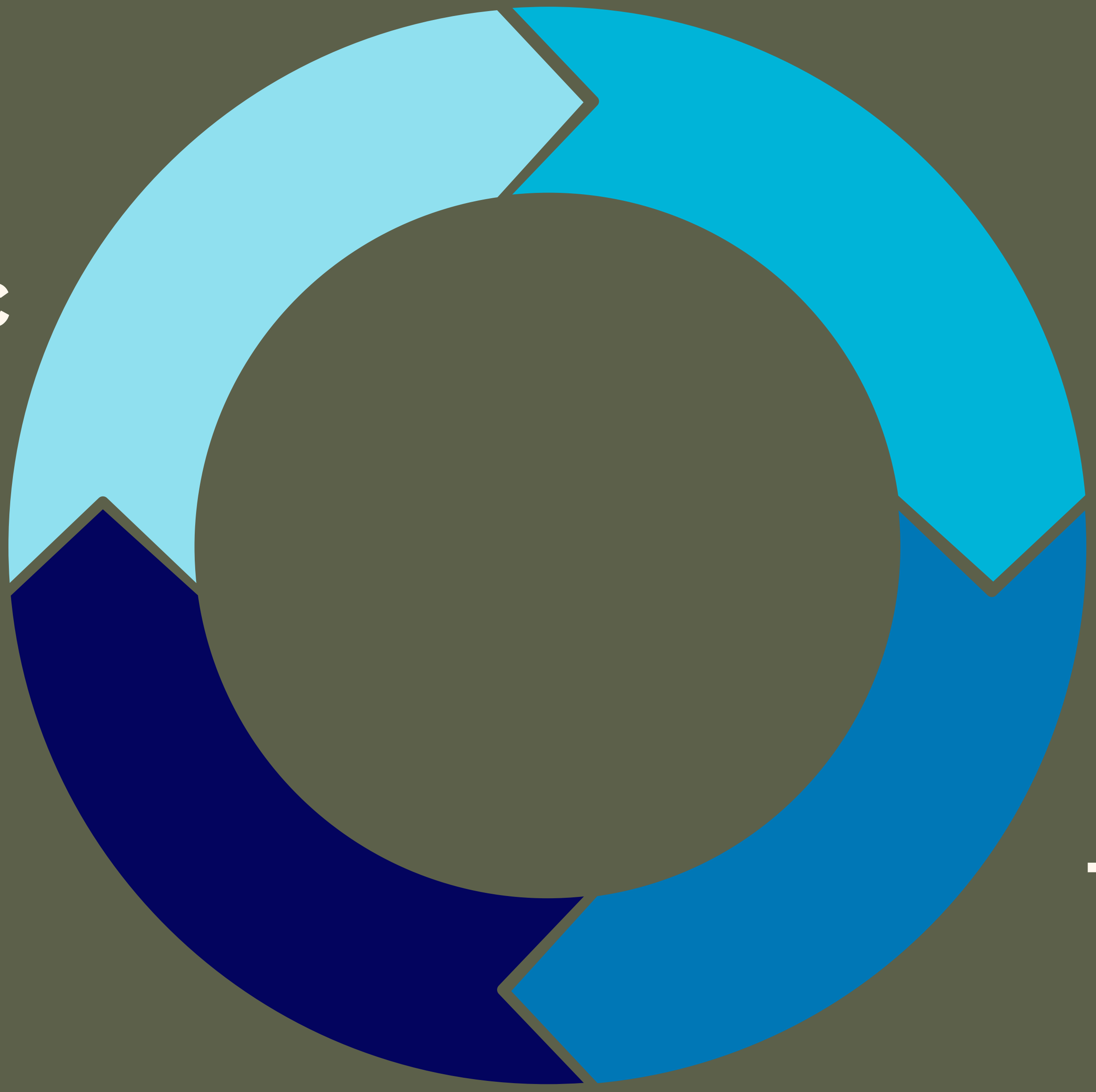


Commodity Prices Spiked



Natural gas (USD/MMBtu) 4.6080 +0.0150 (+0.33%)

Economic Impacts



Gas prices spike

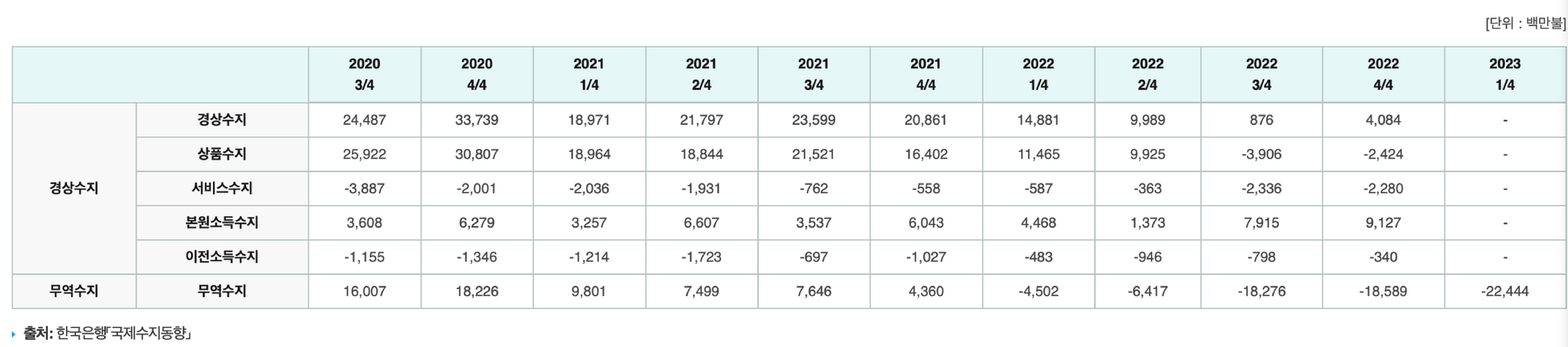
Transportation prices spike

Commodity price spike

Inflation,
Discourages Economic Activity

SK's Economic Impact

1. Balance of Trade



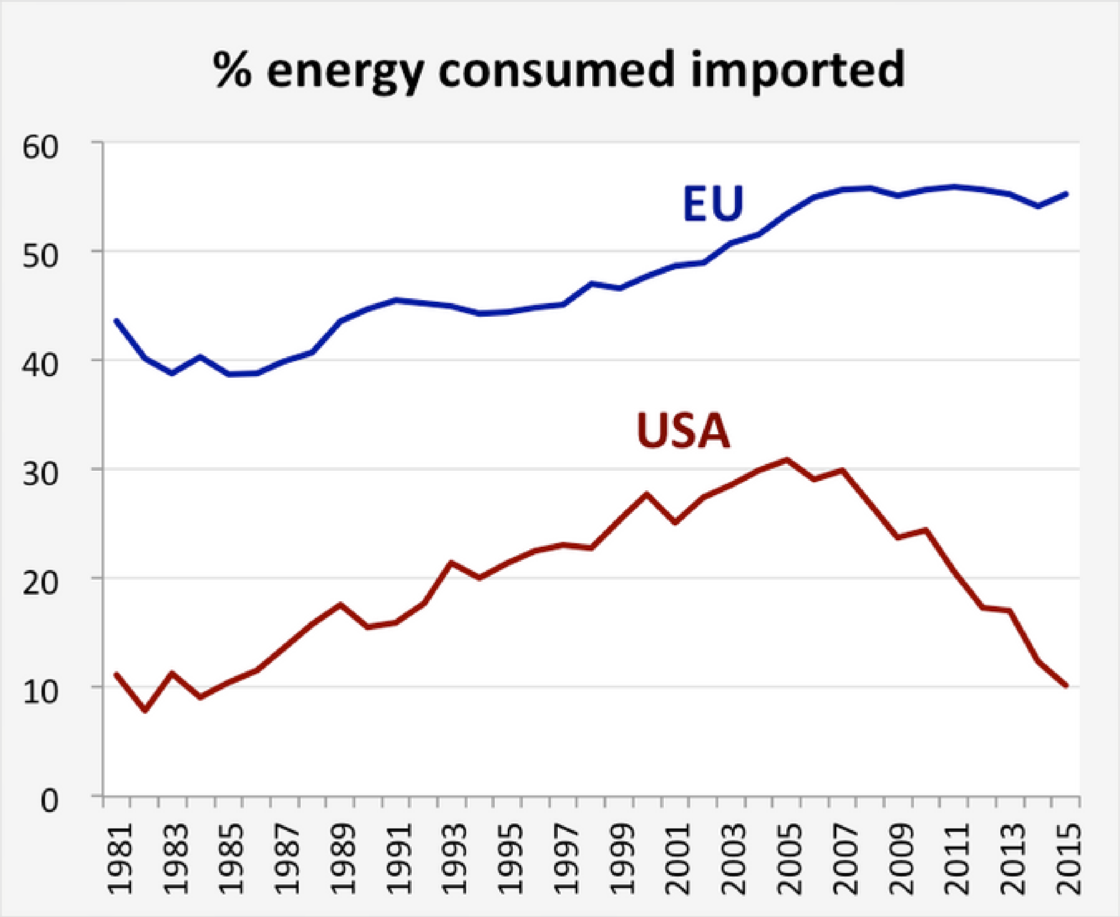
[단위 : 백만불]

		2020 3/4	2020 4/4	2021 1/4	2021 2/4	2021 3/4	2021 4/4	2022 1/4	2022 2/4	2022 3/4	2022 4/4	2023 1/4
경상수지	**경상수지**	24,487	33,739	18,971	21,797	23,599	20,861	14,881	9,989	876	4,084	-
	상품수지	25,922	30,807	18,964	18,844	21,521	16,402	11,465	9,925	-3,906	-2,424	-
	서비스수지	-3,887	-2,001	-2,036	-1,931	-762	-558	-587	-363	-2,336	-2,280	-
	본원소득수지	3,608	6,279	3,257	6,607	3,537	6,043	4,468	1,373	7,915	9,127	-
	이전소득수지	-1,155	-1,346	-1,214	-1,723	-697	-1,027	-483	-946	-798	-340	-
무역수지	**무역수지**	16,007	18,226	9,801	7,499	7,646	4,360	-4,502	-6,417	-18,276	-18,589	-22,444

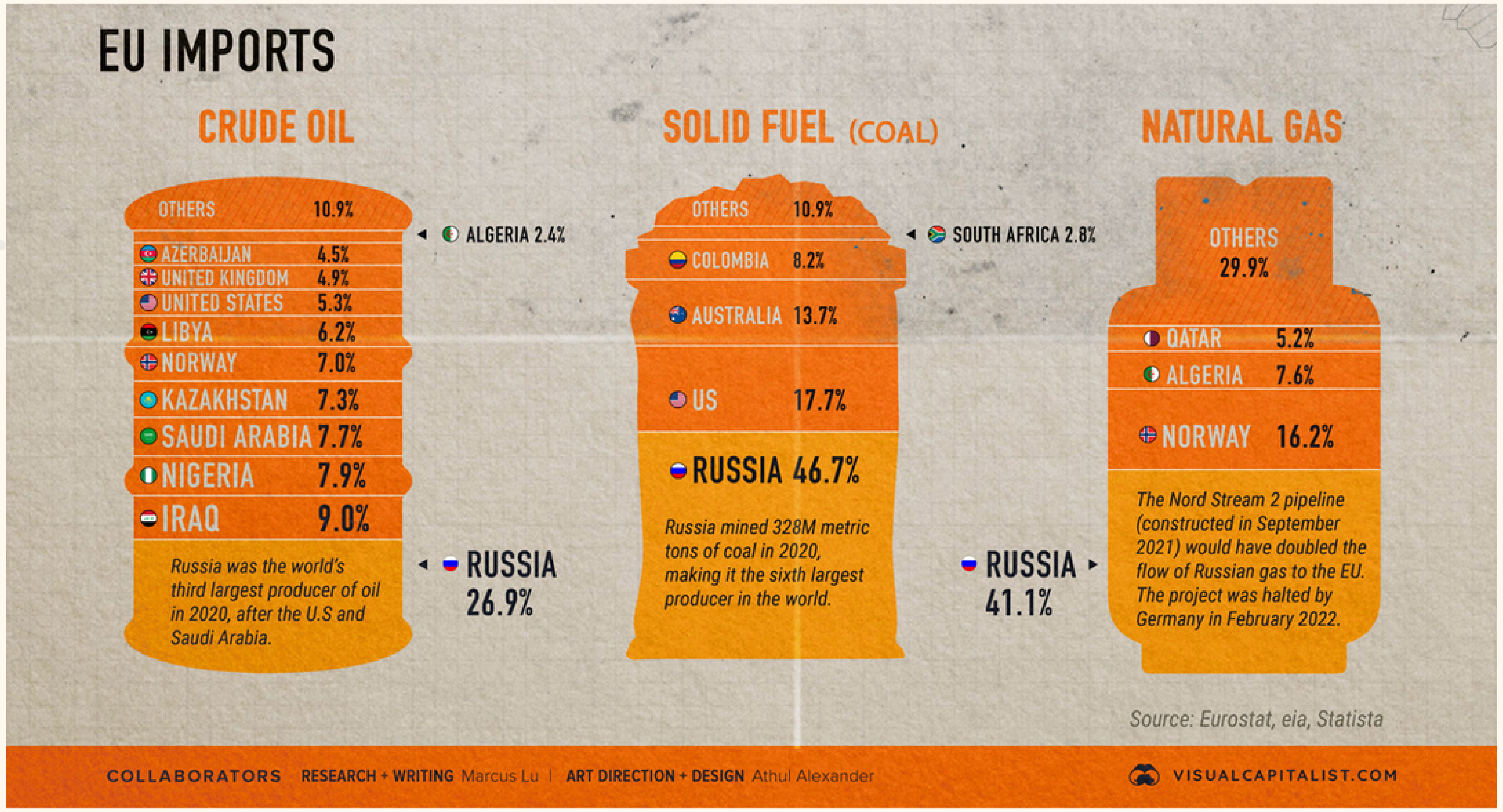
- **출처:** 한국은행「국제수지동향」

Causes of Energy Dependence

- Geographic lottery winners and losers
- Lack of diversification in importers
- Decrease in nuclear energy



% energy consumed imported



EU IMPORTS

Russia was the world's third largest producer of oil in 2020, after the U.S and Saudi Arabia.

Russia mined 328M metric tons of coal in 2020, making it the sixth largest producer in the world.

The Nord Stream 2 pipeline (constructed in September 2021) would have doubled the flow of Russian gas to the EU. The project was halted by Germany in February 2022.

Source: Eurostat, eia, Statista

COLLABORATORS **RESEARCH + WRITING** Marcus Lu | **ART DIRECTION + DESIGN** Athul Alexander

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Solution

1. Self-sufficiency
2. Diversification
    a. between renewable, nuclear energy
    b. include fossil fuel
    c. import from different countries
3. Invest in infrastructure & technology

"New Government Energy Policy Direction"

새정부 에너지정책 방향 공청회

Nuclear energy ratio to a minimum of 30% by 2030

Alleviating monopoly on sale of electricity

Energy supply -> Energy demand efficiency

Energy Security

1. What is the Korean public sentiment toward nuclear energy? And how do think the Korean government should persuade the public to build nuclear power plants in their towns, considering the NIMBY phenomenon?

2. How has the Ukrainian war/Covid 19 triggered inflation affected your personal life?

3. Is renewable energy as eco-friendly as they say? What potential harm does it have and how can we avoid it?

4. Some argue that energy security can only be achieved through self-sufficiency. Others argue the mix with foreign fossil fuels is inevitable. As the South Korean President which of the two would you set as the long-term goal?